x_1	x_2	x_3	x_4	x_5	x_6
1	2	1	5	6	6
1	1	2	4	5	6
C_1	C_1	C_1	C_2	C_2	C_2

Initial $W = \begin{pmatrix} 4 \\ -2 \\ -2 \end{pmatrix}$

REPEAT using Step η = 1

Check classes of $Z_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $Z_2 = \begin{pmatrix} 7 \\ 6 \end{pmatrix}$, $Z_3 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

C_1 positive C_2 negative **Zero** is unaccepted

Ugmented Vectors (from Top)

x_1	x_2	x_3	x_4	x_5	x_6
1	1	1	1	1	1
1	2	1	5	6	6
1	1	2	4	5	6
C_1	C_1	C_1	C_2	C_2	C_2

Iteration 2

No change in W

Stop Iterations

$d(x_1) = \begin{matrix} 1 & 1 & 1 \end{matrix} \begin{pmatrix} 5 \\ -1 \\ -1 \end{pmatrix} = 3$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix} 5 \\ -1 \\ -1 \end{pmatrix}$

$d(x_2) = \begin{matrix} 1 & 2 & 1 \end{matrix} \begin{pmatrix} 5 \\ -1 \\ -1 \end{pmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix} 5 \\ -1 \\ -1 \end{pmatrix}$

$d(x_3) = \begin{matrix} 1 & 1 & 2 \end{matrix} \begin{pmatrix} 5 \\ -1 \\ -1 \end{pmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix} 5 \\ -1 \\ -1 \end{pmatrix}$

$d(x_4) = \begin{matrix} 1 & 5 & 4 \end{matrix} \begin{pmatrix} 5 \\ -1 \\ -1 \end{pmatrix} = -4$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix} 5 \\ -1 \\ -1 \end{pmatrix}$

$d(x_5) = \begin{matrix} 1 & 6 & 5 \end{matrix} \begin{pmatrix} 5 \\ -1 \\ -1 \end{pmatrix} = -6$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix} 5 \\ -1 \\ -1 \end{pmatrix}$

$d(x_6) = \begin{matrix} 1 & 6 & 6 \end{matrix} \begin{pmatrix} 5 \\ -1 \\ -1 \end{pmatrix} = -7$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix} 5 \\ -1 \\ -1 \end{pmatrix}$